

PATHOLOGY

1- CNS Infection

- infections of the CNS are uncommon.
- - Anatomically, infections of the CNS may involve:
 1. Meninges and superficial spaces (meningitis, empyema).
 2. Brain parenchyma (encephalitis, and cerebral abscess).
 3. Ventricles, choroid plexus
- Infections of the CNS may be divided into 3 broad groups:
 - 1. Bacterial**
 - (a) Acute, due to pus forming bacteria.
 - (b) Chronic granulomatous inflammatory response eg. TB.
 - 2. Viral**
 - 3. Miscellaneous**
 - (a) Fungal eg. Cryptococcosis.
 - (b) Protozoal eg. Malaria, toxoplasmosis,.
 - (c) Metazoal eg. Cysticercosis, hydatid cyst.
- The infective agent may reach the CNS by various routes as:
 1. Blood stream.
 2. Direct extension, e.g. Middle ear, sinuses.
 3. Penetrating wounds
- 4. There is often spread via CSF pathways. This results in variations in CSF protein levels, sugar content and presence of inflammatory cells, findings which help to establish the diagnosis of CNS infection

Infection include the following

A-Meningitis

B- cerebral abscess

C-encephalitis

Bacterial Meningitis – Organisms

Less common than viral

- **Aetiology:** Commonly responsible organisms vary with the age of the patient:

-E.coli: neonates.

-Haemophilus influenza: infants & children.

-Neisseria meningitidis: adolescents & young adults.

- Streptococcus Pneumoniae : young & old.

- **Neisseria meningitidis:**
 - it is the typical agents of infectious bacterial meningitis
 - Its classic presentation is as an epidemic infection

Most adolescents do not have protective antibodies against this organism so there is a high incidence of person to person infection

- It causes septicemia and it can be fatal

Older adults tend to have protective antibodies so epidemics infection in these populations do not occur

	Normal	Bacterial	Viral	TB
Appearance	normal	Turbid	Clear	Fibrin web
cells		100,000	<2000	250-500
		polymorph	lymphocyte	lymphocyte
glucose	0.6	Low<0.4	>0.6	<0.4
protein	0.18-0.45	>1 high	<1	
Gram stain		+ve	-ve	

- *Streptococcus pneumoniae*:

- Produce meningitis in susceptible individuals also most of patients do not develop meningitis due to effective blood brain barrier
- Meningitis occurs in debilitated and elderly persons or in young

children

- Untreated cases are usually fatal.
- ***Haemophilus influenza***: it was a very important cause of meningitis in children however widespread vaccination reduce its incidence infants & children.

Bacterial Meningitis – Pathogenesis

First there is Infection of upper respiratory tract followed by Invasion of blood stream (bacteraemia) then Seeding & inflammation of meninges

Viral meningitis

- Most common infection of CNS especially in <1yr and most least damaging form of meningitis
- Causes: enterovirus (commonest, meningitis occurring in 50% of children <3mth) herpes simplex 2, influenza, rubella, Mumps,
- Mononuclear lymphocytes in CSF
- There is headache and other symptoms and signs related to increase of ICP
- Symptomatic treatment most of cases are self-limited.
- Complications associated with encephalitis and ICP

TB meningitis

- Usually insidious: difficult to diagnose in early stages (fever 30%, URTI 20%)
- Rare in children in developed countries
- If untreated is usually fatal
- Meningitis usually occurs 3-6mths after primary infection
- 1 stage-lasts 1-2wk, fever malaise, headache

- 2 stage-+/- suddenly, meningeal signs
- 3 stage-worsening neurological condition, death

Aseptic meningitis Its causes include

- Endocarditis
- Infection with spirochetes
- Malaria
- Fungal (cryptococcal, Neisseria meningitidis)
- Amoebic meningitis

Clinical manifestation

- In adults Severe headache is the common symptoms
- followed by nuchal rigidity
- classic Triad
- Nuchal rigidity
- Sudden high fever
- Altered mental status
- Positive Kernig's sign
- Positive Brudzinksi's sign

Complications:

- 1. Arterial thrombosis →→ Cerebral infarction.
- 2. Thrombophlebitis →→ Pyemia.
- 3. Adrenal haemorrhage due to septicemia.
- 4. Fibrosis.

Cerebral abscess

Caused by bacterial infection however it can result from parasitic infection

Routes of infection direct spread (infection of middle ear, mastoid air space) 2-haematogenous spread

Organism is mixed and frequently anaerobic

it appears as swelling has liquefied centre filled with pus surrounding cerebral oedema is common

Microscopically as any abscess

Complications:

1. Increased intracranial pressure.
2. Rupture of the abscess into ventricles.
3. Spread of infection → meningitis, and thrombophlebitis.
- 4-Cerebral abscess can initiate seizures
- 5-Abscess in epidural space usually occurs along the vertebral column, causing pressure in spinal cord and nerve roots leading to localizing neurological symptoms

Encephalitis

Definition: acute diffuse inflammation affecting brain parenchyma Most commonly viral

Herpes viruses (HSV1, HSV-2, CMV) Enteroviruses (echoviruses)

Paramyxoviruses (measles, mumps)

Herpes simplex can cause severe encephalitis

This virus has a tendency to attack several organs and is unselective in its effect on nervous tissue (pantropic reaction), It is rare

neonatal herpes simplex infection results in a fatal disseminated infection with

encephalitis

Other Diseases A- TB can cause

1- TB meningitis

2- brain abscess which obstruct CSF flow resulting in hydrocephalus

3- TB osteomyelitis of the vertebrae (Pott`s disease) cause inflammation and dysfunction of the spinal nerve roots

B- HIV it can directly infect neurons or it may predisposes to infectious complications in the CNS

C- cytomegalovirus can cause encephalitis in immunosupressed patients.